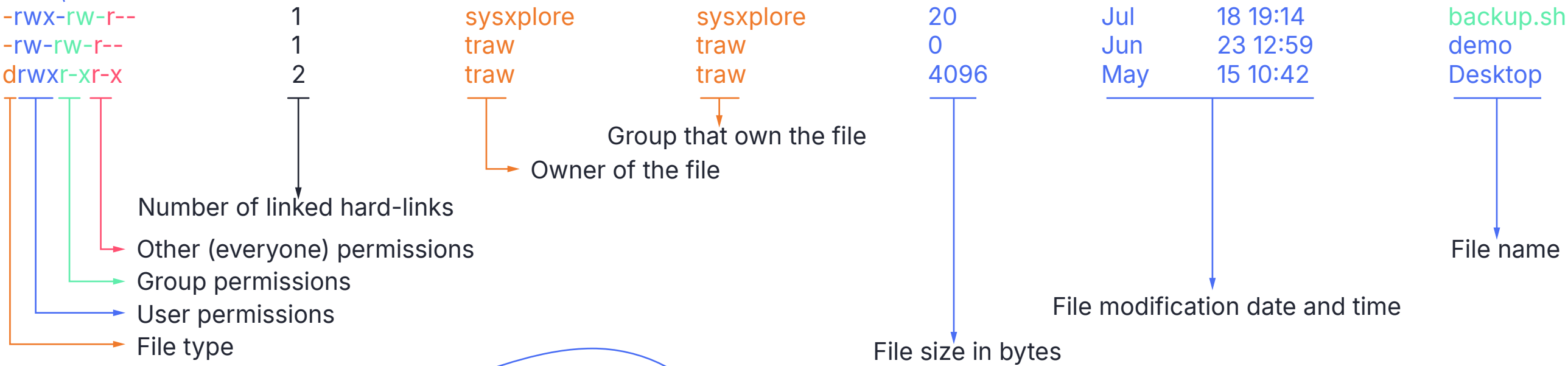


# LINUX FILE PERMISSIONS

sysxplore@zorinos ~\$ ls -l  
total 93792



User Owner Permissions Group Owner Permissions Other Permissions

`-rwxrw-r--`

|   |         |   |   |
|---|---------|---|---|
| r | Read    | 4 | 7 |
| w | Write   | 2 |   |
| x | Execute | 1 |   |

|   |               |   |   |
|---|---------------|---|---|
| r | Read          | 4 | 6 |
| w | Write         | 2 |   |
| - | No permission | 0 |   |

|   |               |   |   |
|---|---------------|---|---|
| r | Read          | 4 | 4 |
| - | No permission | 0 |   |
| - | No permission | 0 |   |

SUID Permission

Group Owner Permissions

User Owner Permissions Other Permissions

`-rwsrw-r--`

\$ chmod u+s file

SGID Permission

Group Owner Permissions

User Owner Permissions Other Permissions

`drwxrwsr--`

\$ chmod g+s directory\_name

Sticky Bit Permission

Group Owner Permissions

User Owner Permissions Other Permissions

`drwxrwxrwt`

\$ chmod +t directory\_name

| Binar | Octal     | Permissions            | Representation |
|-------|-----------|------------------------|----------------|
| y     | 0 (0+0+0) | No Permission          | ---            |
| 000   | 1 (0+0+1) | Execute                | --x            |
| 001   | 2 (0+2+0) | Write                  | -w-            |
| 010   | 3 (0+2+1) | Write + Execute        | -wx            |
| 011   | 4 (4+0+0) | Read                   | r--            |
| 100   | 5 (4+0+1) | Read + Execute         | r-x            |
| 101   | 6 (4+2+0) | Read + Write           | rw-            |
| 110   | 7 (4+2+1) | Read + Write + Execute | rwX            |

| Owner |   |   | Group |   |   | Other |   |   |
|-------|---|---|-------|---|---|-------|---|---|
| r     | w | x | r     | w | x | r     | w | x |
| S     | s |   | S     | s |   | T     | t |   |

@sysxplore

Capital S is an error it occurs if you set SUID bit or SGID bit to a file without execute (x) bit set

Capital T is an error it occurs if you set Sticky bit to a file without execute (x) bit set